A.I. Image Detector

Distinguishing Human vs. AI-Generated Images with Deep Learning

Team 2 · Yasin Ahmed · Kaiye Liu · Tyler Rimkunas

The Problem

70%

of social media images are now AI-generated

78%

of users struggle to tell AI from real

Generative AI has flooded social media with synthetic imagery. The growing inability to distinguish real from fabricated content creates serious risks. These range from misinformation campaigns to fabricated evidence that can incite panic or violence.

Goal: build a binary classifier that correctly labels at least 80% of 11,992 images as AI-generated or human-created.

Our Dataset

- Source: AI vs. Human-Generated Images dataset (Kaggle, Sala et al.)
- Human images sourced from Shutterstock across diverse categories
- Each real image is paired with an AI-generated counterpart
- 79,950 labeled training images: 50% human, 50% AI
- 5,540 unlabeled test images (not used —>see Data Preparation)
- Labels: 0 = human, 1 = AI-generated

	Unnamed: 0	file_name	label
0	0	train_data/a6dcb93f596a43249135678dfcfc17ea.jpg	1
1	1	train_data/041be3153810433ab146bc97d5af505c.jpg	0
2	2	train_data/615df26ce9494e5db2f70e57ce7a3a4f.jpg	1
3	3	train_data/8542fe161d9147be8e835e50c0de39cd.jpg	0
4	4	train_data/5d81fa12bc3b4cea8c94a6700a477cf2.jpg	1

[Figure 1: train.csv]

	id
0	test_data_v2/1a2d9fd3e21b4266aea1f66b30aed157.jpg
1	test_data_v2/ab5df8f441fe4fbf9dc9c6baae699dc7.jpg
2	test_data_v2/eb364dd2dfe34feda0e52466b7ce7956.jpg
3	test_data_v2/f76c2580e9644d85a741a42c6f6b39c0.jpg
4	test_data_v2/a16495c578b7494683805484ca27cf9f.jpg

[Figure 2: test.csv]

Data Preparation

The original unlabeled test CSV was discarded. The 79,950-image training set was split into three directory-structured subsets using a Python script, enabling TensorFlow's image_dataset_from_directory pipeline for efficient GPU loading.

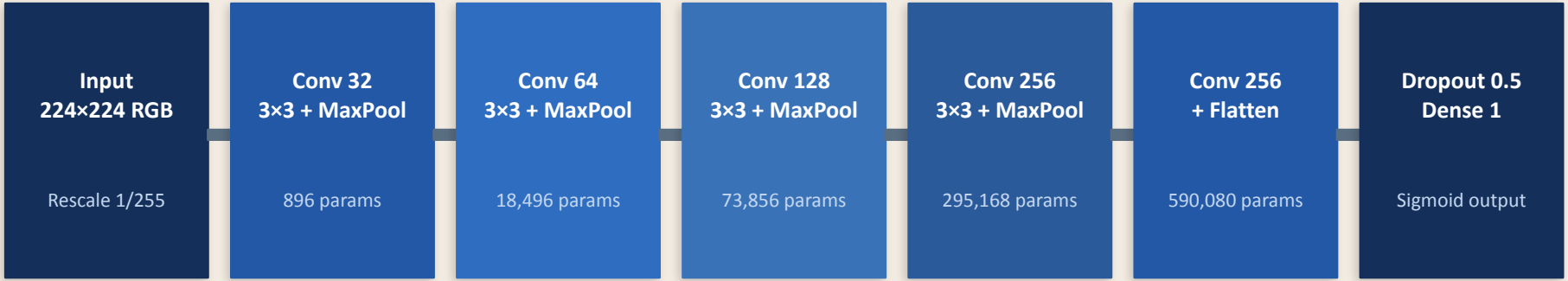
Split	Images	Role
Train	55,965	Model training
Validation	11,993	Monitor overfitting during training
Test	11,992	Held out — final evaluation only

Key decisions

224×224 px	Standard CNN input resolution
Rescaling 1/255	Stable gradient flow
Batch size 32	Balances throughput on A100 GPU
RGB colour	Full colour information retained

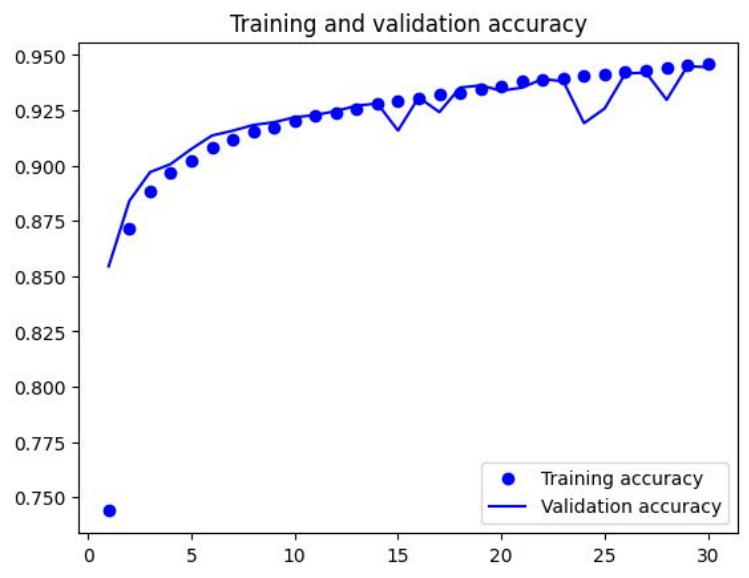
Model Architecture

Five-block Custom CNN · TensorFlow / Keras · 1,004,097 total parameters (3.83 MB)

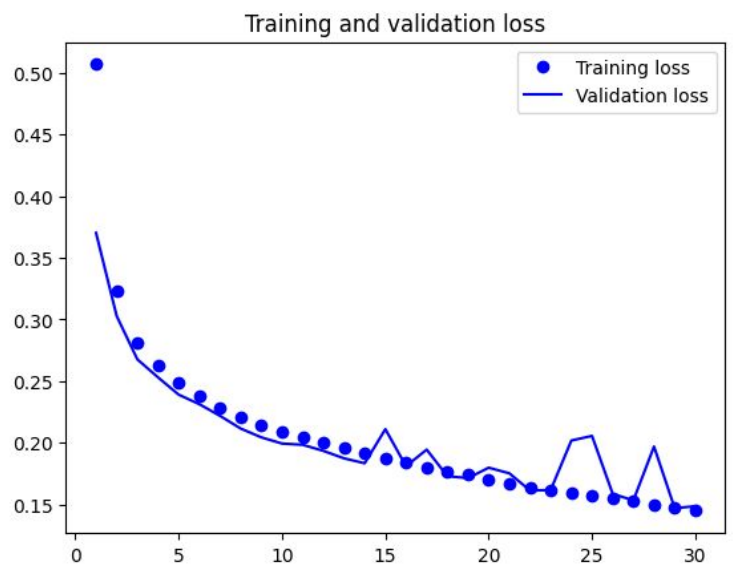


- Progressive filter depth (32→256) learns increasingly abstract features
- MaxPooling after each block reduces spatial dimensions and aids translational invariance
- Dropout (0.5) before the output layer prevents overfitting on the large training set
- RMSprop optimizer at learning rate 1×10^{-5} for stable, conservative convergence
- Binary cross-entropy loss with sigmoid output — probabilities above 0.5 are classified as AI

Training Results



[Figure 3: Training & Validation Accuracy Plot]



[Figure 4: Training & Validation Loss Plot]

Training accuracy rose from 71.8% to 94.6% over 30 epochs. Validation accuracy peaked at 94.86% at epoch 29, which was saved by ModelCheckpoint. The small and consistent gap between training and validation curves confirms that Dropout regularization successfully controlled overfitting throughout.

Evaluation

95.0%

Test Accuracy
(Target was 80%)

Class	Precision	Recall	F1	Support
Human (0)	0.97	0.92	0.94	5,996
AI (1)	0.92	0.97	0.95	5,996
Macro Avg	0.95	0.94	0.94	11,992

vs. SGD Baseline



Error Analysis

	Predicted: Human	Predicted: AI
Actual: Human	TN = 5,510	**FP = 486**
Actual: AI	**FN = 178**	TP = 5,818

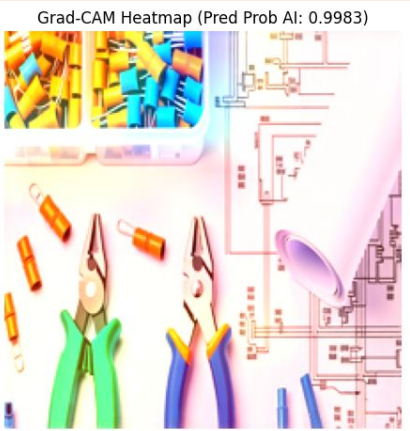
- 486 false positives : mostly **heavily edited** or **digital graphic design** images mistaken for AI
- 178 false negatives : AI images with **realistic shadows** and **fine facial detail**

The model tends to flag AI more conservatively, a reasonable bias for real-world content moderation where false positives carryhigher cost.

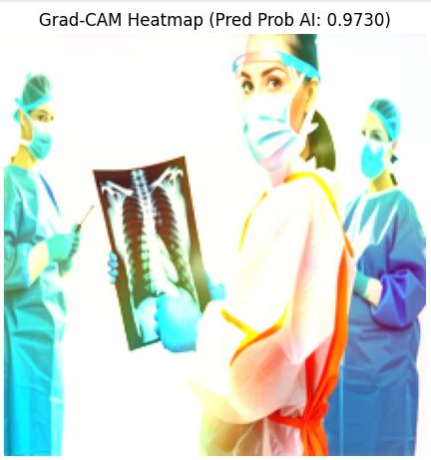
Example: TRUE POSITIVE

A.I. generated images correctly identified

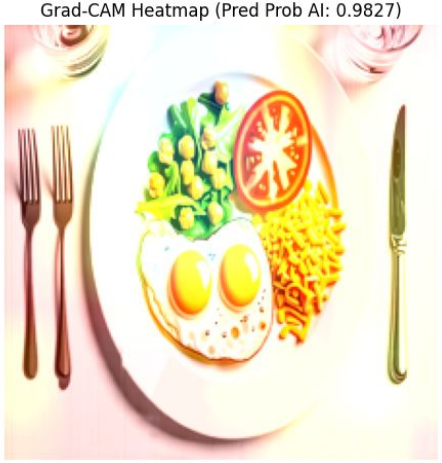






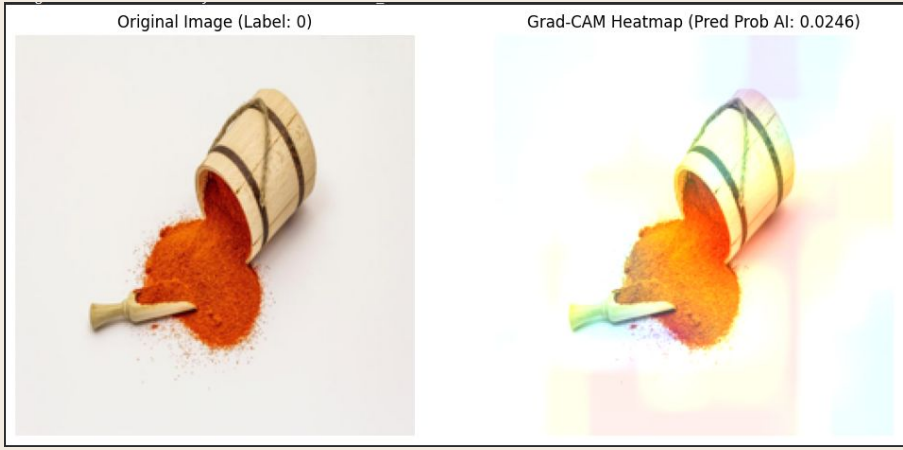


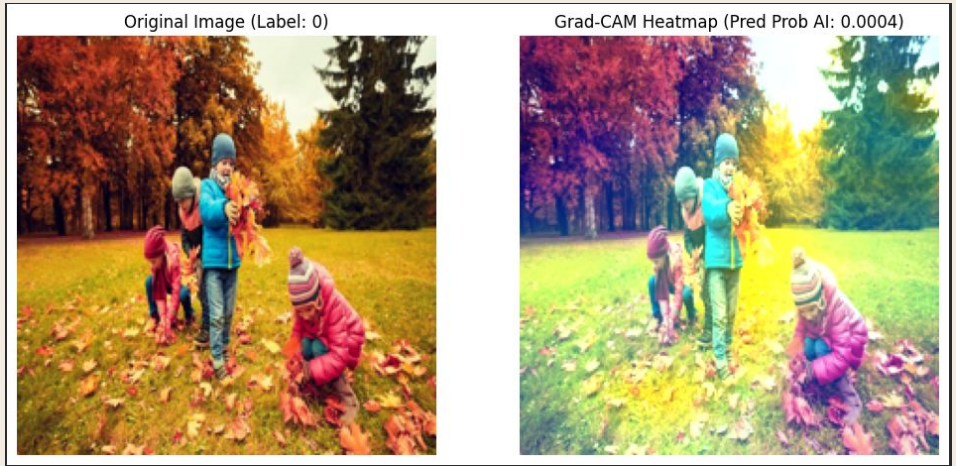


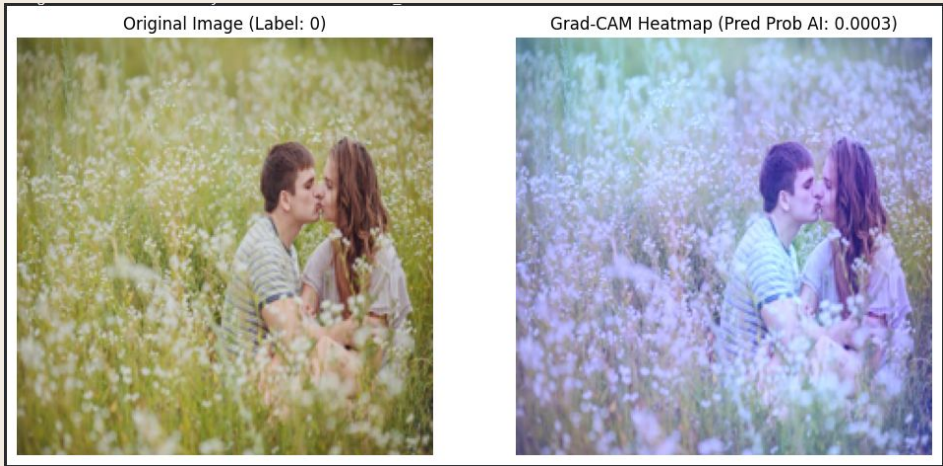


Example: TRUE NEGATIVE

Human made images
correctly identified







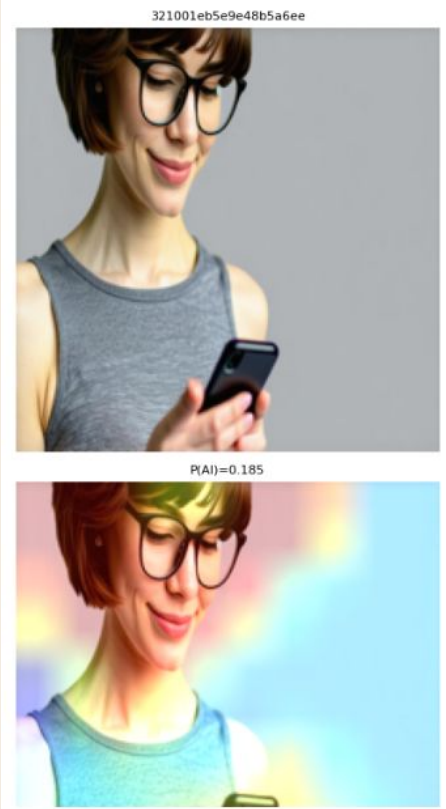
Example: FALSE POSITIVES



Human made graphic / digital design

Mistaken for AI generated content

Example: FALSE NEGATIVES



AI generated images with good shadows and facial details

Predicted to be Human/real image

How We Got There

RESOLUTION	28×28 grayscale pixels → flattened vector	**224×224 RGB → directory-based TF pipeline**
ARCHITECTURE	SGD classifier (scikit-learn, no GPU)	**Five-block CNN on A100 GPU (Google Colab Pro)**
REGULARIZATION	No regularization: overfitting risk	**Dropout (0.5) + ModelCheckpoint at best val loss**
EXPLAINABILITY	Black-box predictions only	**Grad-CAM visualizations for model interpretability**

Conclusions

95.0%

Overall test accuracy got well beyond our 80% target

0.97

Precision for human images

0.97

Recall for AI-generated images

The model learned to detect subtle pixel-level artifacts inherent in AI-generated images, as confirmed by Grad-CAM analysis. CNNs trained at scale are viable tools for digital content verification.

Future Work

- Fine-tune ResNet50 / EfficientNet / Vision Transformer to push beyond 97%
- Add data augmentation (flips, rotations, JPEG artifacts) for real-world robustness
- Test generalization on newer generators (Stable Diffusion, Midjourney v6)
- Explore frequency-domain features to catch artifacts invisible in pixel space
- Deploy as a browser extension or web API for content moderation pipelines